

E-Commerce Sales Analysis & BI Dashboard Project Report

1. Project Overview

This project analyzes e-commerce sales data to generate business insights using Python, ML, and Power BI.

2. Objectives

To analyze sales trends, customer behavior, and build predictive machine learning models.

3. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Power BI.

4. Dataset

Customers, Products, Orders, Payments datasets with merged final dataset.

5. Feature Engineering

TotalSales, DiscountAmount, FinalAmount, Profit, Profit Margin.

6. Analysis Performed

EDA, correlation analysis, pivot tables, time series analysis, hypothesis testing.

7. Machine Learning

Linear Regression model used to predict sales with good accuracy ($R^2 \sim 0.86$).

8. Dashboard

Power BI dashboard includes KPIs, trends, categories, and customer insights.

9. Conclusion

End-to-end BI pipeline successfully implemented for business decision making.